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(1)

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Semester : 5
Subject : Stat ST-101

Assignment No 01

Question No 01:

Raw Data:

91, 124, 109, 129, 141, 107, 86, 76, 118, 111, 99, 99, 114, 100,
88, 108, 87, 107, 101, 71, 63, 121, 122, 111, 119, 77, 127, 61,
133, 68, 77, 177, 110, 95, 96, 96, 88, 106, 119, 79, 81, 127,
86, 153, 79, 129, 151, 89, 143, 147, 90, 142, 105, 94, 125, 96, 99,
138, 145, 113, 129, 87, 113, 110, 144, 91, 106, 104, 97, 115, 100, 117,
73, 134, 108, 102, 123, 106, 119, 104, 101, 120, 112, 138, 140, 103,
96, 136, 78, 83, 75, 100, 113, 114, 109, 116, 109, 116, 104, 128

Range : $X_m = 177$, $X_o = 61$

Range = $X_m - X_o = 177 - 61 = 116$

No. of classes = $1 + 3 \cdot 3 \lg(N) = 1 + 3 \cdot 3 \lg(100)$
 $= 1 + 3 \cdot 3 \lg(100) = 1 + 8.6 = 7.6 \approx 8$

Class Interval (h) = $\frac{R}{m} = \frac{116}{8} = 14.5 \approx 15$

But as in question, class interval is provided which is 5, so, we will use that:

Grouped frequency Distribution Table is as follows:

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Class limits	Tally Marks	No. of commodities (f)
61-65		2
66-70		1
71-75		3
76-80		6
81-85		2
86-90		8
91-95		4
96-100		11
101-105		9
106-110		11
111-115		9
116-120		8
121-125		5
126-130		6
131-135		2
136-140		4
141-145		5
146-150		1
151-155		2
156-160		
161-165		
166-170		
171-175		
176-180		1
		100 = Σf

Question No 02:

(i)

Solution:

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Days fewer than 4;

Means: $X < 4$

f_i for $X_i = 0, 1, 2, 3$

$$\text{Sum} = \text{Days} = 5 + 7 + 9 + 6 = (27)$$

(ii)

Solution:

Days greater or equal (at least) 4;

Means: $X \geq 4$

f_i for $X_i = 4, 5, 6, 7$

$$\text{Sum} = 4 + 2 + 1 + 1 = (8)$$

(iii)

Solution:

X	f	fX
0	5	0
1	7	7
2	9	18
3	6	18
4	4	16
5	2	10
6	1	6
7	1	7
	35	82

Total Number of absences over 35 days =
 $\sum fX = 82$

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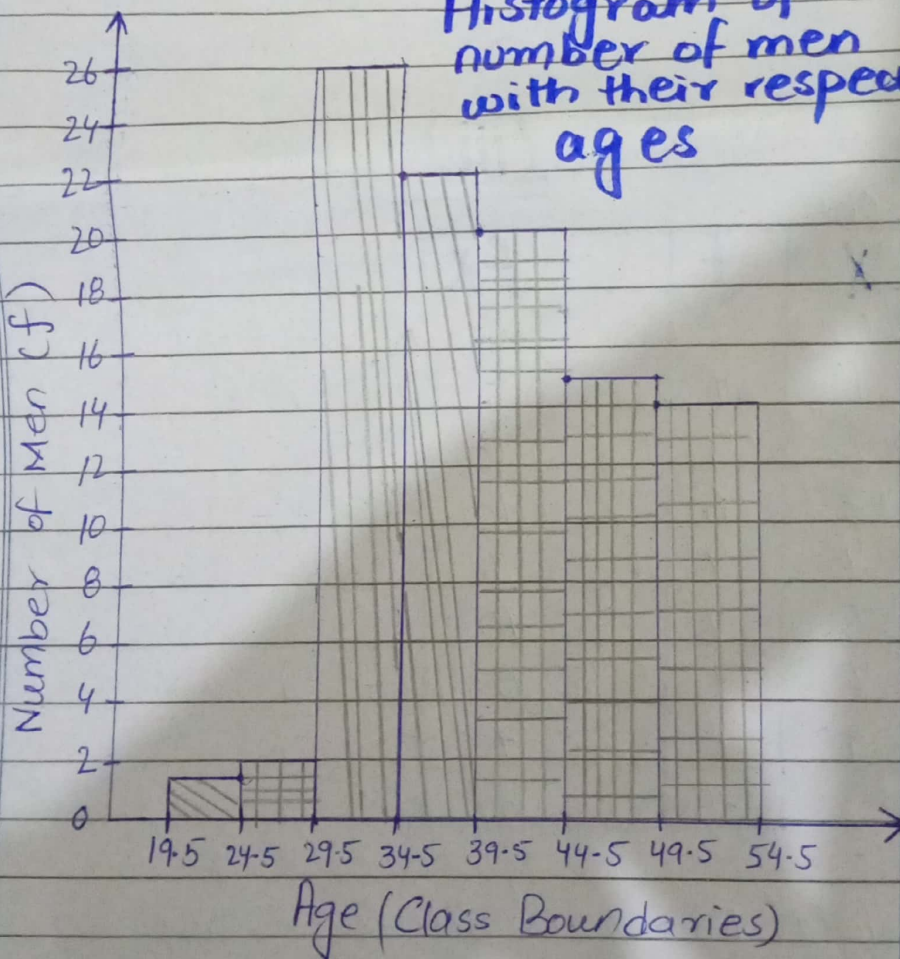
Question No 03:

Solution:

for Histogram, we need class boundaries

Age (class limit)	No. of ^{Men} (f)	Age (Class Boundaries)
20-24	1	19.5-24.5
25-29	2	24.5-29.5
30-34	26	29.5-34.5
35-39	22	34.5-39.5
40-44	20	39.5-44.5
45-49	15	44.5-49.5
50-54	14	49.5-54.5

Histogram of
number of men
with their respective
ages



Question No 04:

Solution:

Class limits	freq.	Relative freq.	Cumulative freq.	Cumulative %
8-12	5	0.25	5	25
13-17	1	0.05	6	25+5=30
18-22	3	0.15	9	30+15=45
23-27	6	0.30	15	45+30=75
28-32	5	0.25	15+5=20	75+25=100
	20	1.00		

(i) As we know for class limits, difference / class interval (h) is same for all classes, $h=(5)$ is the class interval, so filling accordingly

(ii) finding frequency of class "23-27"
if we do $(15 - 9)$ of this class c.f and its previous c-f we get freq. of this class $15 - 9 = 6$

(iii) for finding Σf , we can use relative frequency formulae which is:

$$\frac{\text{freq. (of that class)}}{\Sigma f} = \text{R-f (of that class)}$$

class "23-27"

$$\frac{6}{A = \Sigma f} = 0.30 \Rightarrow A = \frac{6}{0.30} = (20) = \Sigma f$$

(iv) As we know R-f total = 1.00 and c-f last row is always equal to Σf , from that we get to know freq. of last class.

(v) find relative freq. of last class.

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- (vi) C.P last row is always 100%.
- (vii) for class "13-17", frequency can be find

$$\frac{f}{\Sigma f} = R \cdot f \Rightarrow \frac{f}{20} = 0.05$$

$$f = 0.05 \times 20 = 1$$
- (viii) All others will be filled accordingly.

Question No 05:

Solution:-

30, 5, 29, 7, 20, 31, 28, 10, 13, 24, 25, 7, 22, 15, 15, 22,
 78, 79, 99, 30, 80, 80, 75, 63, 24, 35, 25, 90, 35, 33,
 70, 63, 35, 17, 9

(i)

Stem (leading digit)	leaf (trailing digit)	Ordered values
0	5, 7, 7, 9	5, 7, 7, 9
1	0, 3, 5, 5, 7	0, 3, 5, 5, 7
2	9, 0, 8, 4, 5, 2, 2, 4, 5	0, 2, 2, 4, 4, 5, 5, 8, 9
3	0, 1, 0, 5, 5, 3, 5	0, 0, 1, 3, 5, 5, 5
4		
5		
6	3, 3	3, 3
7	8, 9, 5, 0	0, 5, 8, 9
8	0, 0	0, 0
9	9, 0	0, 9

Order Arranged Data:

5, 7, 7, 9, 10, 13, 15, 15, 17, 20, 22, 22, 24, 24, 25, 25,
 28, 29, 30, 30, 31, 33, 35, 35, 35, 63, 63, 70, 75, 78, 79,
 80, 80, 90, 99

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(ii)

Stem	leaf	Ordered values
0-2	5, 7, 9, 0, 8, 0, 3, 4, 5, 7, 2, 5, 5, 2, 4, 5, 7, 9	5, 7, 7, 9, 0, 3, 5, 5, 7, 9, 2, 2, 4, 4, 5, 5, 8, 9
3-5	0, 1, 0, 5, 5, 3, 5	0, 0, 1, 3, 5, 5, 5
6-9	8, 9, 9, 0, 0, 5, 3, 0, 0, 3	3, 3, 0, 5, 8, 9, 0, 0, 0, 9

Arranged Data:

5, 7, 7, 9, 10, 13, 15, 15, 17, 20, 22, 22, 24, 24, 25, 25,
28, 29, 30, 30, 31, 33, 35, 35, 63, 63, 70, 75, 78, 79,
80, 80, 90, 99

Question No 06:

Solution:

As it is Ungroup Data, we will use formulae of ungroup data.

X	$\log(X)$	$\left(\frac{1}{X}\right)$	$\left(\frac{1}{X}\right)$ of ungroup data.
60	1.7782	0.01667	0.0167
80	1.9031	0.0125	0.0125
90	1.9542	0.0111	0.0111
96	1.9823	0.0104	0.0104
120	2.0792	0.8333×10^{-4}	0.0083
150	2.1761	0.6667×10^{-4}	0.0067
200	2.3010	0.5×10^{-4}	0.0050
360	2.5563	0.2778×10^{-4}	0.0028
480	2.6812	0.2083×10^{-4}	0.0021
520	2.7160	0.1923×10^{-4}	0.0019
1060	3.0253	0.9434×10^{-4}	0.0009
1200	3.0792	0.8333×10^{-5}	0.0008
1450	3.1614 3.1614	0.6897×10^{-5}	0.0007
2500	3.3979	0.4×10^{-5}	0.0004
7200	3.8574	0.1389×10^{-5}	0.0001
15,566	3.86487	0.2011	0.00072

Arithmetic Mean:

As we know:

$$A.M = \bar{X} = \frac{\sum X}{n} = \frac{15,566}{15}$$

$$\bar{X} = 1037.7333$$

Geometric Mean:

As we know:

$$G.M = \text{Antilog} \left[\frac{\sum \log(X)}{n} \right]$$

$$= \text{Antilog} \left[\frac{38.6487}{15} \right] \Rightarrow \text{Antilog}(2.5766)$$

$$G.M = 377.2246$$

Harmonic Mean:

As we know:

$$H.M = \frac{n}{\sum \left(\frac{1}{X} \right)} = \frac{15}{0.0804}$$

$$H.M = 186.5672$$

$$H.M = 186.5672$$

Question No 09:

Solution:

(i)

Arithmetic Mean:

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(3)

Weekly income	No of workers (f)	X	FX	log X	flog X	($\frac{f}{x}$)
35-39	15	37	555	1.5682	23.5230	0.4054
40-44	13	42	546	1.6232	21.1022	0.3095
45-49	17	47	799	1.6721	28.4257	0.3617
50-54	29	52	1,508	1.7160	49.7641	0.5577
55-59	11	57	627	1.7559	19.3146	0.1930
60-64	10	62	620	1.7924	17.9239	0.1629
65-69	6	67	402	1.8261	10.9564	0.0896
	101		5,057		171.0099	2.0798

As we know:

$$AM = \bar{X} = \frac{\sum FX}{\sum f} = \frac{5,057}{101}$$

$$\bar{X} = 50.0693$$

(ii)

Geometric Mean:

As we know:

$$G.M = \text{Antilog} \left[\frac{\sum f \log X}{\sum f} \right]$$

$$= \text{Antilog} \left[\frac{171.0099}{101} \right]$$

$$= \text{Antilog} (1.6932)$$

$$G.M = 49.3364$$

(iii)

Harmonic Mean:

As we know:

$$H.M = \frac{\sum f}{\sum \left(\frac{f}{x} \right)} = \frac{101}{2.0798} = 48.5624$$

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(iv)Median:

As we know:

$$\bar{x} = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

C-B	f	C-f	
34.5-39.5	15	15	
39.5-44.5	13	28	$\left(\frac{n}{4}\right)$
44.5-49.5	17	45	
49.5-54.5	29	74	$\left(\frac{n}{2}\right), \left(\frac{7n}{10}\right)$
54.5-59.5	11	85	$\left(\frac{3n}{4}\right)$
59.5-64.5	10	95	$\left(\frac{85n}{100}\right)$
64.5-69.5	6	101	
	101		

$$\frac{n}{2} = \frac{101}{2} = 50.5 \text{ lies in "49.5-54.5"}$$

Here, $l = 49.5$, $h = 5$, $f = 29$, $n = 101$, $c = 45$

$$\bar{x} = 49.5 + \frac{5}{29} \left(\frac{101}{2} - 45 \right)$$

$$= 49.5 + \frac{5}{29} (50.5 - 45)$$

$$= 49.5 + \frac{5(5.5)}{29}$$

$$= 49.5 + \frac{27.5}{29}$$

$$= 49.5 + 0.9483$$

$$\boxed{\bar{x} = 50.4483}$$

(v)Q₁ & Q₃:

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$$Q_1 = l + h \left(\frac{1n}{f} - c \right)$$

$$\frac{n}{4} = \frac{101}{4} = 25.25 \text{ lies in " } 39.5 - 44.5 \text{ "}$$

$$\text{Here, } l = 39.5, h = 5, f = 13, n = 101, c = 15$$

$$Q_1 = 39.5 + \frac{5}{13} \left(\frac{101}{4} - 15 \right)$$

$$= 39.5 + \frac{5}{13} (25.25 - 15)$$

$$= 39.5 + \frac{5}{13} (10.25)$$

$$= 39.5 + \frac{51.25}{13}$$

$$= 39.5 + 3.9423$$

$$Q_1 = 43.4423$$

$$Q_3 = l + h \left(\frac{3n}{4} - c \right)$$

$$\frac{3n}{4} = \frac{3 \times 101}{4} = 75.75 \text{ lies in " } 54.5 - 59.5 \text{ "}$$

$$\text{Here, } l = 54.5, h = 5, f = 11, n = 101, c = 74$$

$$Q_3 = 54.5 + \frac{5}{11} \left(\frac{3 \times 101}{4} - 74 \right)$$

$$= 54.5 + \frac{5}{11} (75.75 - 74)$$

$$= 54.5 + \frac{5(1.75)}{11}$$

$$= 54.5 + \frac{8.75}{11}$$

$$= 54.5 + 0.7955$$

$$Q_3 = 55.2955$$

(vi)Mode:

As we know:

$$\hat{X} = l + \left[\frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \right] \times h$$

$f = 29$ as highest freq. lies in "49.5-54.5"

Here, $l = 49.5$, $f_m = 29$, $f_1 = 17$, $f_2 = 11$, $h = 5$

$$\hat{X} = 49.5 + \left[\frac{(29 - 17)}{(29 - 17) + (29 - 11)} \right] \times 5$$

$$= 49.5 + \left[\frac{12}{12 + 18} \right] \times 5$$

$$= 49.5 + \frac{60}{30}$$

$$= 49.5 + 2$$

$$\boxed{\hat{X} = 51.5}$$

(vii) D_7 :

As we know:

$$D_7 = l + \frac{h}{f} \left(\frac{7n}{10} - c \right)$$

$$\frac{7n}{10} = \frac{7 \times 101}{10} = 70.7 \text{ lies in "49.5-54.5"}$$

Here, $l = 49.5$, $h = 5$, $f = 29$, $n = 101$, $c = 45$

$$D_7 = 49.5 + \frac{5}{29} \left(\frac{7 \times 101}{10} - 45 \right)$$

$$= 49.5 + \frac{5}{29} (70.7 - 45)$$

$$= 49.5 + \frac{5}{29} (25.7)$$

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(4)

$$= 49.5 + \frac{128.5}{29}$$

$$= 49.5 + 4.4310$$

$$[D_7 = 53.9310]$$

(viii)P₈₅:

As we know:

$$P_{85} = l + \frac{h}{f} \left(\frac{85n}{100} - c \right)$$

$$\frac{85n}{100} = \frac{85 \times 101}{100} = 85.85 \text{ lies in "59.5-64.5"}$$

$$\text{Here, } l = 59.5, h = 5, f = 10, n = 101, c = 85$$

$$P_{85} = 59.5 + \frac{5}{10} \left(\frac{85 \times 101}{100} - 85 \right)$$

$$= 59.5 + \frac{5}{10} (85.85 - 85)$$

$$= 59.5 + \frac{5}{10} (0.85)$$

$$= 59.5 + \frac{4.25}{10}$$

$$= 59.5 + 0.425$$

$$[P_{85} = 59.925]$$

(ix)P₉₆:

As we know:

$$P_{96} = l + \frac{h}{f} \left(\frac{96n}{100} - c \right)$$

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$$\frac{96n}{100} = \frac{96 \times 101}{100} = 96.96 \text{ lies in "64.5-69.5"}$$

Here, $l = 64.5$, $h = 5$, $f = 6$, $n = 101$, $c = 95$

$$P_{96} = 64.5 + \frac{5}{6} \left(\frac{96 \times 101}{100} - 95 \right)$$

$$= 64.5 + \frac{5}{6} (96.96 - 95)$$

$$= 64.5 + \frac{5}{6} (1.96)$$

$$= 64.5 + \frac{0.8}{6}$$

$$= 64.5 + 1.6333$$

$$P_{96} = 66.1333$$

(x)

Median as proved:

$$\rightarrow \bar{X} = 50.4483$$

$$\rightarrow Q_2 = l + \frac{h}{f} \left(\frac{2n}{4} - c \right) \Rightarrow Q_2 = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$\frac{n}{2} = \frac{101}{2} = 50.5 \text{ lies in "49.5-54.5"}$$

Here, $l = 49.5$, $h = 5$, $f = 29$, $n = 101$, $c = 45$

$$Q_2 = 49.5 + \frac{5}{29} \left(\frac{101}{2} - 45 \right)$$

$$= 49.5 + \frac{5}{29} (50.5 - 45)$$

$$= 49.5 + \frac{5}{29} (5.5)$$

$$= 49.5 + \frac{27.5}{29}$$

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$$= 49.5 + 0.9483$$

$$[Q_2 = 50.4483]$$

$$D_5 = l + \frac{h}{f} \left(\frac{5n}{10} - c \right)$$

$$\frac{5n}{10} = \frac{5 \times 101}{10} = 50.5 \text{ lies in "49.5-54.5"}$$

Here, $l = 49.5, h = 5, f = 29, n = 101, c = 45$

$$D_5 = 49.5 + \frac{5}{29} \left(\frac{5 \times 101}{10} - 45 \right)$$

$$= 49.5 + \frac{5}{29} (50.5 - 45)$$

$$= 49.5 + \frac{5(5.5)}{29}$$

$$= 49.5 + \frac{27.5}{29}$$

$$= 49.5 + 0.9483$$

$$[D_5 = 50.4483]$$

$$P_{50} = l + \frac{h}{f} \left(\frac{50n}{100} - c \right)$$

$$\frac{50n}{100} = \frac{50 \times 101}{100} = 50.5 \text{ lies in "49.5-54.5"}$$

Here, $l = 49.5, h = 5, f = 29, n = 101, c = 45$

$$P_{50} = 49.5 + \frac{5}{29} \left(\frac{50 \times 101}{100} - 45 \right)$$

$$= 49.5 + \frac{5}{29} (50.5 - 45)$$

$$= 49.5 + \frac{5(5.5)}{29}$$

$$= 49.5 + \frac{27.5}{29}$$

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$$= 49.5 + 0.9483$$

$$P_{50} = 50.4483$$

So As proved

$$\bar{X} = Q_2 = D_5 = P_{50}$$

Question No 10:

Solution:

Boys	Girls
$n_1 = 50$	$n_2 = 40$
$\bar{X}_1 = 59.5$	$\bar{X}_2 = 54.0$
$S_1 = 8.38$	$S_2 = 8.23$

$$\bar{X}_c = ? , S_c = ?$$

As we know:

$$\bar{X}_c = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$

$$= \frac{(50)(59.5) + (40)(54.0)}{50 + 40}$$

$$= \frac{2975 + 2160}{90}$$

$$= \frac{5135}{90}$$

$$\bar{X}_c = 57.0556$$

$$S_c = \sqrt{\frac{n_1 (S_1^2 + d_1^2) + n_2 (S_2^2 + d_2^2)}{n_1 + n_2}}$$

where $d_1 = \bar{X}_1 - \bar{X}_c$ & $d_2 = \bar{X}_2 - \bar{X}_c$

$$d_1 = 59.5 - 57.1 = 2.4$$

$$d_2 = 54.0 - 57.1 = -3.1$$

(5)

$$S_c = \sqrt{\frac{(50)[(8.38)^2 + (2.4)^2] + (40)[(8.23)^2 + (-3.1)^2]}{50 + 40}}$$

$$= \sqrt{\frac{50(75.9844) + 40(77.3429)}{90}}$$

$$= \sqrt{\frac{3799.22 + 3093.716}{90}}$$

$$= \sqrt{\frac{6892.936}{90}}$$

$$= \sqrt{76.5882}$$

$$S_c = 8.75$$

Question No 11:

Solution:

Group 1	Group 2	Group 3
$n_1 = 200$	$n_2 = 250$	$n_3 = 300$
$\bar{X}_1 = 25$	$\bar{X}_2 = 10$	$\bar{X}_3 = 15$
$S_1 = 3$	$S_2 = 4$	$S_3 = 5$

prove:

$$\bar{X}_c = 16, S_c = 7.2, \text{ \& \uparrow find C.V = ?}$$

$$\bar{X}_c = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2 + n_3 \bar{X}_3}{n_1 + n_2 + n_3}$$

$$= \frac{(200)(25) + (250)(10) + (300)(15)}{200 + 250 + 300}$$

$$= \frac{12,000}{750}$$

$$\boxed{\bar{X}_c = 16}, \text{ it is proved that } \boxed{\bar{X}_c = 16}$$

$$S_c = \sqrt{\frac{n_1(S_1^2 + d_1^2) + n_2(S_2^2 + d_2^2) + n_3(S_3^2 + d_3^2)}{n_1 + n_2 + n_3}}$$

where: $d_1 = \bar{X}_1 - \bar{X}_c$, $d_2 = \bar{X}_2 - \bar{X}_c$, $d_3 = \bar{X}_3 - \bar{X}_c$

$$d_1 = 25 - 16 = 9$$

$$d_2 = 10 - 16 = -6$$

$$d_3 = 15 - 16 = -1$$

$$S_c = \sqrt{\frac{200[(3)^2 + (9)^2] + 250[(4)^2 + (-6)^2] + 300[(5)^2 + (-1)^2]}{200 + 250 + 300}}$$

$$= \sqrt{\frac{18,000 + 13,000 + 7,800}{750}}$$

$$= \sqrt{\frac{38,800}{750}} \Rightarrow \sqrt{51.7333}$$

$$S_c = 7.19 \approx 7.2$$

it is proved that $S_c = 7.2$

$$C.V = \frac{S_c}{\bar{X}_c} \times 100$$

$$= \frac{7.2}{16} \times 100$$

$$C.V = 0.45 \times 100 = 45\%$$

Question No 14:

solution:

Ungroup Data:

80, 75, 42, 63, 65, 43, 78, 96, 82, 58, 79, 72, 67,
75, 68

(i)

Range:

$$\text{Range} = X_m - X_o$$

$$X_o = 42, X_m = 96$$

$$\text{Range} = 96 - 42 = \boxed{54}$$

(ii)

Coeff. of Range:

$$\text{Coeff of Range} = \frac{X_m - X_o}{X_m + X_o} = \frac{96 - 42}{96 + 42} = \frac{54}{138} = \boxed{0.39}$$

(iii)

Quartile Deviation:

$$Q.D = (Q_3 - Q_1) / 2$$

$$Q_1 = 1 \left(\frac{n+1}{4} \right)^{\text{th}} \text{value in Arranged Data}$$

Arranged Data:

42, 43, 58, 63, 65, 67, 68, 72, 75, 75, 78, 79, 80, 82, 96

$$Q_1 = \left(\frac{15+1}{4} \right)^{\text{th}} \text{value} \Rightarrow \left(\frac{16}{4} \right)^{\text{th}} \text{value}$$

$$= (4)^{\text{th}} \text{value}$$

$$\boxed{Q_1 = 63}$$

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{\text{th}} \text{value in arranged Data}$$

$$= 3 \left(\frac{15+1}{4} \right)^{\text{th}} \text{value} \Rightarrow \left(\frac{48}{4} \right)^{\text{th}} \text{value}$$

$$= (12)^{\text{th}} \text{value} = \boxed{79}$$

$$= (11)^{\text{th}} \text{value} + 0.5 \left((12)^{\text{th}} \text{value} - (11)^{\text{th}} \text{value} \right)$$

$$= 78 + 0.5(79 - 78)$$

$$= 78 + 0.5(1)$$

$$= 78 + 0.5$$

$$\boxed{Q_3 = 78.5}$$

$$Q.D = \frac{(Q_3 - Q_1)}{2} = \frac{79 - 63}{2} = \frac{16}{2} = 8$$

(iv)
Coeff. of Q.D:

$$\begin{aligned} \text{Coeff of Q.D} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{79 - 63}{79 + 63} \\ &= \frac{16}{142} = 0.11 \end{aligned}$$

(v)
M-D from $\bar{X}$, $\tilde{X}$, $\hat{X}$:

As we know:

$$\text{M-D from Mean} = \frac{\sum |X - \bar{X}|}{n}$$

$$\text{M-D from Median} = \frac{\sum |X - \tilde{X}|}{n}$$

$$\text{M-D from Mode} = \frac{\sum |X - \hat{X}|}{n}$$

for that first we will find $\bar{X}$, $\tilde{X}$, $\hat{X}$:

$$\bar{X} = \frac{\sum X}{n} = \frac{1043}{15} = 69.53$$

As $n=15$, so $\tilde{X}$ = odd formulae:

$$\begin{aligned} \tilde{X} &= \left(\frac{n+1}{2} \right)^{\text{th}} \text{value in arranged Data} = \left(\frac{15+1}{2} \right)^{\text{th}} \text{value} \\ &= \left(\frac{16}{2} \right)^{\text{th}} \text{value} = (8)^{\text{th}} \text{value} \end{aligned}$$

$$\tilde{X} = 72$$

As 75 in data exist two time so;

$$\hat{X} = 75$$

X	$ x - \bar{x} $	$ x - \tilde{x} $	$ x - \hat{x} $	X^2
42	+27.53	+30	+33	1764
43	+26.53	+29	+32	1849
58	+11.53	+14	+17	3364
63	+6.53	+9	+12	3969
65	+4.53	+7	+10	4225
67	+2.53	+5	+8	4489
68	+1.53	+4	+7	4624
72	2.47	0	+3	5184
75	5.47	3	0	5625
75	5.47	3	0	5625
78	8.47	6	3	6084
79	9.47	7	4	6241
80	10.47	8	5	6400
82	12.47	10	7	6724
96	26.47	24	21	9216
1043	161.47	159	162	75383

$$M.D \text{ from } \bar{x} = \frac{161.47}{15} = 10.76$$

$$M.D \text{ from } \tilde{x} = \frac{159}{15} = 10.60$$

$$M.D \text{ from } \hat{x} = \frac{162}{15} = 10.80$$

(vi)
Coeff. of M.D using $\bar{x}$, $\tilde{x}$, $\hat{x}$:

$$\begin{aligned} \text{Coeff. of M.D from Mean} &= \frac{M.D \text{ from Mean}}{\text{Mean}} = \frac{10.76}{69.53} \\ &= 0.1548 \end{aligned}$$

$$\text{Coeff. of M-D} = \frac{\text{M.D from Median}}{\text{Median}} = \frac{10.60}{72} = 0.1472$$

$$\text{Coeff. of M-D} = \frac{\text{M.D from Mode}}{\text{Mode}} = \frac{10.80}{75} = 0.1440$$

(vii) Standard Deviation:

$$\begin{aligned} S_x &= \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} \\ &= \sqrt{\frac{75,383}{15} - \left(\frac{1,043}{15}\right)^2} \\ &= \sqrt{5025.5333 - 4834.8844} \\ &= \sqrt{190.6489} \\ S_x &= 13.81 \end{aligned}$$

(viii) Coeff. of Variation:

$$C.V = \frac{S_x}{\bar{x}} \times 100 = \frac{13.81}{69.53} \times 100 = 19.86\%$$

(ix) Pearson Coeff. of Skewness:

$$S_k = \frac{\text{Mean} - \text{Mode}}{S} = \frac{69.53 - 75}{13.81} = -5.47$$

$$S_k =$$

Bowley's

$$S_k =$$

Here,

$$S_k =$$

$$S_k$$

Quest

Solution

Age

15-19

20-24

25-29

30-34

35-40

41-44

45-49

50-54

Ra

Re

he

R

$$S_k = -0.40$$

Bowley's coeff. of skewness:

$$S_k = \frac{Q_3 + Q_1 - 2 \text{ Median}}{Q_3 - Q_1}$$

Here, $Q_1 = 63$, $Q_3 = 79$, $\bar{X} = 72$

$$S_k = \frac{63 + 79 - 2(72)}{79 - 63} = \frac{-2}{15.5}$$

$$S_k = -0.13$$

Question No 15:

Solution:

Age	No. of men (f)	X	FX	C-B	C-F	FX ²
15-19	29	17	493	14.5-19.5	29	8,381
20-24	176	22	3,872	19.5-24.5	205	85,184
25-29	208	27	5,616	24.5-29.5	413	151,632
30-34	173	32	5,536	29.5-34.5	586	177,152
35-40	82	37	3,034	34.5-39.5	668	112,258
41-44	40	42	1,680	39.5-44.5	708	70,560
45-49	15	47	705	44.5-49.5	723	33,135
50-54	3	52	156	49.5-54.5	726	8,112
	726		21,092			646,414

(i)

Range:

$$\text{Range} = X_m - X_o$$

here $X_m = 52$, $X_o = 17$ {Mid points}

$$\text{Range} = 52 - 17 = 35$$

(ii)
Coeff. of Range:

$$\frac{\text{Coeff of Range}}{\text{Range}} = \frac{X_m - X_o}{X_m + X_o} = \frac{52 - 17}{52 + 17} = \frac{35}{69} = \boxed{0.51}$$

(iii)
Quartile Deviation:

$$Q.D = \frac{(Q_3 - Q_1)}{2}$$

$$Q_1 = l + \frac{h}{f} \left(\frac{n}{4} - c \right)$$

$$\frac{n}{4} = \frac{726}{4} = 181.5 \text{ lies in "19.5 - 24.5"}$$

$$\text{here, } l = 19.5, h = 5, f = 176, n = 726, c = 29$$

$$Q_1 = 19.5 + \frac{5}{176} \left(\frac{726}{4} - 29 \right)$$

$$= 19.5 + \frac{5}{176} (181.5 - 29)$$

$$= 19.5 + \frac{5}{176} (152.5)$$

$$= 19.5 + \frac{762.5}{176}$$

$$= 19.5 + 4.33$$

$$\boxed{Q_1 = 23.83}$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

$$\frac{3n}{4} = \frac{3 \times 726}{4} = 544.5 \text{ lies in "29.5 - 34.5"}$$

$$\text{Here, } l = 29.5, h = 5, f = 173, n = 726, c = 413$$

(7)

$$\begin{aligned}
 Q_3 &= 29.5 + \frac{5}{173} \left(\frac{3 \times 126}{4} - 413 \right) \\
 &= 29.5 + \frac{5}{173} (544.5 - 413) \\
 &= 29.5 + \frac{5 \times 131.5}{173} \\
 &= 29.5 + \frac{657.5}{173} \\
 &= 29.5 + 3.80
 \end{aligned}$$

$$Q_3 = 33.30$$

$$Q.D = \frac{33.30 - 23.83}{2}$$

$$= \frac{9.47}{2}$$

$$Q.D = 4.735 \approx 4.74$$

(iv)

Coeff of Q.D:Coeff of
Q.D

$$\begin{aligned}
 &= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{33.30 - 23.83}{33.30 + 23.83} \\
 &= \frac{9.47}{57.13} = 0.1658
 \end{aligned}$$

(v)

M.D from $\bar{X}$, $\tilde{X}$, $\hat{X}$:

$$M.D \text{ from } \bar{X} = \frac{\sum f |X - \bar{X}|}{\sum f} = \frac{4,035}{726} = 5.56$$

$$M.D \text{ from } \tilde{X} = \frac{\sum f |X - \tilde{X}|}{\sum f} = \frac{3,960}{726} = 5.45$$

$$M.D \text{ from } \hat{X} = \frac{\sum f |X - \hat{X}|}{\sum f} = \frac{3,864.76}{726} = 5.32$$

Age (c)	f	x	fX	C.f	$ x - \bar{x} $	$f x - \bar{x} $	$ x - \tilde{x} $	$f x - \tilde{x} $	$ x - \hat{x} $	$f x - \hat{x} $
14.5-19.5	29	17	493	29	+12.05	+349.45	+11.3	+327.7	+9.89	+286.81
19.5-24.5	176	22	3872	205	+7.05	+1240.8	+6.3	+1108.8	+4.89	+860.64
24.5-29.5	208	27	5616	413	+2.05	+426.4	+1.3	+270.4	0.11	22.88
29.5-34.5	173	32	5536	586	-2.95	-510.35	-3.7	-640.1	-5.11	-884.03
34.5-39.5	82	37	3034	668	-7.95	-651.9	-8.7	-713.4	-10.11	-829.02
39.5-44.5	40	42	1680	708	-12.95	-518	-13.7	-548	-15.11	-604.4
44.5-49.5	15	47	705	723	-17.95	-269.25	-18.7	-280.5	-20.11	-301.65
49.5-54.5	3	52	156	726	-22.95	-68.85	-23.7	-71.7	-25.11	-75.33
	726		21092			4,035		3,960		3864.76

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{21,092}{726} = \boxed{29.05}$$

$$\tilde{X} = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

Here; $\frac{n}{2} = \frac{726}{2} = 363$ lies in "24.5-29.5"

Here; $l = 24.5, h = 5, f = 208, n = 726, c = 205$

$$\tilde{X} = 24.5 + \frac{5}{208} \left(\frac{726}{2} - 205 \right)$$

$$= 24.5 + \frac{5}{208} (363 - 205)$$

$$= 24.5 + \frac{5 \times 158}{208}$$

$$= 24.5 + \frac{790}{208}$$

$$= 24.5 + 3.80$$

$$\boxed{\tilde{X} = 28.30}$$

$$\hat{X} = l + \left[\frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \right] \times h$$

$\hat{X}$ lies in "24.5 - 29.5"

$l = 24.5, f_m = 208, f_1 = 176, f_2 = 173, h = 5$

$$\begin{aligned}
 \hat{x} &= 24.5 + \left[\frac{(208-176)}{(208-176)+(208-173)} \right] \times 5 \\
 &= 24.5 + \frac{32 \times 5}{32+35} \\
 &= 24.5 + \frac{160}{67} \\
 &= 24.5 + 2.39 \\
 \boxed{\hat{x} = 26.89}
 \end{aligned}$$

(vi)
Coeff of M.D using $\bar{x}$, $\tilde{x}$, $\hat{x}$:

$$\begin{aligned}
 \text{Coeff of M.D using Mean} &= \frac{\text{M.D from } \bar{x}}{\bar{x}} = \frac{5.56}{29.05} \\
 &= \boxed{0.1914}
 \end{aligned}$$

$$\begin{aligned}
 \text{Coeff of M.D using Median} &= \frac{\text{M.D from } \tilde{x}}{\tilde{x}} = \frac{5.45}{28.36} = \boxed{0.1926}
 \end{aligned}$$

$$\begin{aligned}
 \text{Coeff of M.D using Mode} &= \frac{\text{M.D from } \hat{x}}{\hat{x}} = \frac{5.32}{26.89} = \boxed{0.1978}
 \end{aligned}$$

(vii)
Standard Deviation:

$$\begin{aligned}
 S_x &= \sqrt{\frac{\sum fX^2}{\sum f} - \left(\frac{\sum fX}{\sum f} \right)^2} \\
 &= \sqrt{\frac{646,414}{726} - \left(\frac{21,092}{726} \right)^2} \\
 &= \sqrt{890.38 - 844.04} \\
 &= \sqrt{46.34} \\
 \boxed{S_x = 6.81}
 \end{aligned}$$

(viii)
Coeff. of Variation:

$$C.V = \frac{\sum x}{\bar{x}} \times 100 = \frac{6.81}{29.05} \times 100$$
$$= \boxed{23.44\%}$$

(ix)
Pearson's Coeff. of Skewness:

$$S_k = b_1 = \frac{\text{Mean} - \text{Mode}}{S}$$
$$= \frac{29.05 - 26.89}{6.81} = \frac{2.16}{6.81} = \boxed{0.317}$$

(x)
Bowley's Coeff. of Skewness:

$$S_k = \frac{Q_3 + Q_1 - 2 \text{Median}}{Q_3 - Q_1}$$
$$= \frac{23.83 + 33.30 - 2(28.30)}{33.30 - 23.83}$$
$$= \frac{57.13 - 56.6}{9.47}$$
$$= \frac{0.53}{9.47}$$
$$\boxed{S_k = 0.06}$$

Question No 12:

(8)

Gp	f	X	fX	(x- $\bar{x}$)	(x- $\bar{x}$) ²	f(x- $\bar{x}$)	(x- $\bar{x}$) ³	f(x- $\bar{x}$) ³	(x- $\bar{x}$) ⁴	f(x- $\bar{x}$) ⁴
2-4	18	3	54	-7.76	60.22	1083.92	-467.31	-8411.53	3626.48	65276.07
4-6	24	5	120	-5.76	33.18	796.26	-191.12	-4586.80	1100.91	26421.90
6-8	47	7	329	-3.76	14.14	664.47	-53.17	-2498.82	199.94	9397.16
8-10	80	9	720	-1.76	3.10	247.81	-5.46	-436.48	9.61	768.8
10-12	102	11	1,122	0.24	0.06	5.88	0.01	1.47	3.6 $\times 10^{-3}$	0.37
12-14	66	13	858	2.24	5.02	331.16	11.24	742.16	25.20	151.20
14-16	40	15	600	4.24	17.98	719.10	76.24	3049.41	323.28	12931.22
16-18	21	17	357	6.24	38.94	817.69	242.99	5102.70	1516.32	31842.80
18-20	15	19	285	8.24	67.90	1018.46	559.50	8392.44	4610.41	69156.15
	413		4,445			5684.75		1354.55		215945.67

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{4,445}{413} = 10.76$$

$$m_1 = \frac{\sum f(x-\bar{x})^1}{\sum f}$$

$$m_0 = \frac{\sum f(x-\bar{x})^0}{\sum f} = \frac{\sum (1)}{\sum f} = \frac{\sum f}{\sum f} = 1$$

$$m_1 = \frac{\sum f(x-\bar{x})^1}{\sum f} = \frac{0}{\sum f} = 0$$

$$\therefore \sum f(x-\bar{x}) = 0$$

$$m_2 = \frac{\sum f(x-\bar{x})^2}{\sum f} = \frac{5684.75}{413} = 13.76$$

$$m_3 = \frac{\sum f(x-\bar{x})^3}{\sum f} = \frac{1354.55}{413} = 3.28$$

$$b_1 = \frac{(m_3)^2}{(m_2)^3} = \frac{(3.28)^2}{(13.76)^3} = \frac{10.76}{2605.29} = 4.1300 \times 10^{-3}$$

$$b_1 = 0.0041$$

$$b_2 = \frac{m_4}{(m_2)^2} = \frac{522.87}{(13.76)^2} = \boxed{2.76}$$

It is a platykurtic Distribution.

$$m_4 = \frac{\sum f(x-\bar{x})^4}{\sum f} = \frac{215945.67}{413} = \boxed{522.87}$$

Question No 13:

$$X = 10 + 5U \Rightarrow U = \frac{(X-10)}{5}$$

$$\sum f = 125, \sum fU = -46, \sum fU^2 = 806, \\ \sum fU^3 = -242, \sum fU^4 = 1962$$

→ $\bar{X} = ?$ here $A = 10$ & $h = 5$

$$\begin{aligned} \bar{X} &= A + \left(\frac{\sum fU}{\sum f} \right) \times h \\ &= 10 + \left(\frac{-46}{125} \right) \times 5 \\ &= 10 + \left(\frac{-230}{125} \right) \\ &= 10 - 1.84 \\ &= \boxed{\bar{X} = 8.16} \end{aligned}$$

→ Variance (S_x^2) = ?

$$\begin{aligned} S^2 &= \left[\frac{\sum fU^2}{\sum f} - \left(\frac{\sum fU}{\sum f} \right)^2 \right] \times h^2 \\ &= \left[\frac{806}{125} - \left(\frac{-46}{125} \right)^2 \right] \times 5^2 \\ &= (6.448 - 0.1354) \times 25 \\ &= (6.31) \times 25 \\ &= \boxed{S^2 = 157.75} \end{aligned}$$

$$b_1 = \frac{(m_3)^2}{(m_2)^3} = ?$$

$$\text{here } m_r = h^r \frac{\sum f u^r}{\sum f}$$

$$\text{so, } m_3 = h^3 \left(\frac{\sum f u^3}{\sum f} \right) = 5^3 \left(\frac{-242}{125} \right) \\ = 125 \left(\frac{-242}{125} \right) = +125(-1.936)$$

$$\boxed{m_3 = -242}$$

$$m_2 = h^2 \times \frac{\sum f u^2}{\sum f} = 5^2 \left(\frac{806}{125} \right) \\ = 125(6.448)$$

$$\boxed{m_2 = 161.2}$$

$$b_1 = \frac{(-242)^2}{(161.2)^3} = \frac{58564}{4188852.928} = \boxed{0.0140}$$

$$b_2 = \frac{m_4}{(m_2)^2} = ?$$

$$m_4 = h^4 \frac{\sum f u^4}{\sum f} = 5^4 \left(\frac{1962}{125} \right) \\ = 625(15.696)$$

$$\boxed{m_4 = 9810}$$

$$b_2 = \frac{9810}{(161.2)^2} = \frac{9810}{25985.44} = \boxed{0.3775}$$

for a distribution to be normal

$b_2 = 3$ but here $b_2 \neq 3$ so distribution is not normal.

Question No 08:

As we want to find average speed, we will use harmonic mean.

$$H.M = \frac{n}{\sum(\frac{1}{x})} = \frac{10}{0.4433} = \boxed{22.5581}$$

X	($\frac{1}{x}$)
10	0.1
15	0.0667
20	0.05
25	0.04
20	0.05
30	0.0333
40	0.025
50	0.02
30	0.0333
40	0.025
	0.4433

Average speed of Bus is 22.5581.

Question No 07:

For calculating percentage annual increase, we calculate it through geometric mean.

$$\text{For first year (10\%)} = \frac{10}{100} = 0.1$$

$$\text{For second year (20\%)} = \frac{20}{100} = 0.2$$

$$\text{For third year (25\%)} = \frac{25}{100} = 0.25$$

$$\text{Annual increase} = G.M = (0.1 \times 0.2 \times 0.25)^{1/3} = \boxed{0.1710}$$

As we want Annual % increase

$$= 0.1710 \times 100$$

$$\text{Annual \% increase} = \boxed{17.10\%}$$